Expression test: soluble and insoluble fractions (SDS-PAGE)

What you're going to get / What you expect

Paired total and soluble fractions resolved by SDS-PAGE, so overexpression (seen in the total fraction) can be told apart from correct folding (seen in the soluble fraction).

Materials list (per sample (1 mL of culture at OD600 3))

- Induced culture.
- 50 mM sodium phosphate buffer.
- Sonicator (probe); refrigerated centrifuge.
- SDS-PAGE gels; 5X loading buffer; heat block at 95 ºC.
- Prestained protein standard (e.g. NZYBlue).

Steps

1. Collect biomass equivalent to 1 mL of culture at OD600 3 and centrifuge at 1,900 x g for 10 minutes at 4 C.
2. Discard the supernatant and resuspend the pellet in 300 uL of 50 mM sodium phosphate buffer.
3. Disrupt by sonication on ice: a 30 s pulse at 80% amplitude, then 5 minutes on ice to dissipate the heat, then a final 30 s pulse at equal amplitude.
4. Reserve a 30 uL aliquot of the crude lysate as the total fraction.
5. Centrifuge the remaining lysate at 18,300 x g for 10 minutes at 4 ºC and reserve a 30 uL aliquot of the supernatant as the soluble fraction.
6. Mix each 30 uL aliquot with 7.5 uL of 5X loading buffer, denature at 95 ºC for 5 minutes, and load alongside the prestained standard; run the gel.

Troubleshooting

- A band present in the total fraction but absent from the soluble fraction means inclusion bodies: proceed to the solubilisation strategies.
- Keep the lysate cold and pulse with rest intervals to avoid heat denaturation during sonication.